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Artificial Intelligence-Supported Solfège Instruction in Higher Music Education: Effects on Student Performance and Learning Attitudes

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Abstract

Background: Artificial intelligence (AI)-supported learning environments are increasingly used in music education; however, evidence regarding their effectiveness in solfège instruction remains limited. This action research study evaluated the effects of AI-supported solfège instruction on undergraduate students' attitudes and performance. Materials and Methods: This action research study included 36 undergraduate students enrolled in a conservatory program. A 10-week AI-supported solfège training was implemented using the EarMaster intelligent tutoring system. Data were collected through a solfège attitude scale and a performance test administered before and after the intervention, along with a delayed retention test. Results: Following the intervention, significant improvements were observed in both attitude scores (104.6 vs. 117.3, $p < 0.001$) and performance scores (32.8 vs. 51.52, $p < 0.001$). However, retention test scores showed a significant decline after a no-practice period (51.5 vs. 48.8, $p < 0.001$). Qualitative findings indicated increased motivation, engagement, and individualized learning opportunities. Conclusions: AI-supported solfège instruction improved students' performance and learning attitudes in higher music education. The findings suggest that adaptive feedback, individualized practice, and continuous engagement may contribute positively to auditory skill development and student motivation. However, sustained practice remains necessary for long-term retention. Artificial intelligence-supported systems should therefore be integrated as complementary tools alongside teacher-guided instruction to support more flexible and personalized learning environments.

Keywords: artificial intelligence; music education; solfège; EarMaster; intelligent tutoring systems; performance; attitude; learning analytics

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1. Introduction

The rapid advancement of digital technologies has profoundly transformed educational systems, necessitating the integration of innovative instructional approaches to meet the demands of contemporary learners. In recent decades, traditional teacher-centered paradigms have gradually shifted toward learner-centered models that emphasize active participation, individualized learning, and technological integration [1–3]. This transformation has been further accelerated by global developments such as the COVID-19 pandemic, which compelled educational institutions to adopt technology-driven learning environments and highlighted the limitations of conventional instructional methods [4,5]. As a result, the incorporation of emerging technologies into education is no longer optional but essential for maintaining educational quality and relevance in the digital age [6].

Among these emerging technologies, artificial intelligence (AI) has gained increasing attention due to its potential to revolutionize teaching and learning processes. AI-based systems, particularly intelligent tutoring systems (ITS), provide adaptive, personalized, and interactive learning experiences by simulating human instructional behaviors and offering real-time feedback [7–9]. These systems are grounded in cognitive learning theories and are designed to tailor instructional content according to individual learner needs, thereby enhancing both learning efficiency and engagement [10,11]. In educational contexts, AI has been shown to improve academic performance, facilitate self-regulated learning, and support individualized instruction across various disciplines [3,12–17].

In the context of music education, the type of artificial intelligence used in instructional systems is particularly important. The present study focuses on an intelligent tutoring system (ITS)-based learning environment that utilizes adaptive feedback mechanisms, performance monitoring, and individualized task adjustment rather than generative artificial intelligence models. EarMaster operates primarily through rule-based and adaptive instructional algorithms that analyze learner responses, provide immediate corrective feedback, monitor accuracy and progression levels, and dynamically adjust exercise difficulty according to individual performance patterns [18–20]. Such systems are designed to support self-regulated learning and repetitive skill acquisition, both of which are essential components of auditory and performance-based music education.

From a pedagogical perspective, artificial intelligence-supported learning environments are not merely technological tools but components of broader teaching and learning systems that promote learner-centered, adaptive, and self-regulated instruction. Such systems are grounded in contemporary educational theories emphasizing active participation, individualized feedback, reflective learning, and continuous skill development. In music education, where auditory perception, repetitive practice, and performance-based learning are central, pedagogical models that combine instructor guidance with adaptive digital support may provide more flexible and responsive learning environments. Therefore, the present study was designed not only to evaluate the effectiveness of an AI-supported tool, but also to examine its role within a pedagogically structured teaching and learning process in higher music education.

Music education, as a domain that combines theoretical knowledge and practical skill development, has also been significantly influenced by technological advancements. Solfège training, a fundamental component of music education, plays a critical role in developing auditory perception, pitch recognition, rhythm comprehension, and sight-reading skills [21–23]. However, traditional solfège instruction is often constrained by limited instructional time, large class sizes, and insufficient opportunities for individualized feedback, which may hinder student progress and reduce motivation [24,25]. These challenges are particularly evident in higher education settings, where diverse student abilities require differentiated instructional approaches that are difficult to implement in conventional classroom environments.

In this context, AI-supported music education tools have emerged as promising solutions to address these limitations [26–28]. Applications such as EarMaster, which are based on intelligent tutoring principles, enable students to engage in individualized, self-paced learning while receiving immediate and continuous feedback on their performance [29–31]. These systems incorporate adaptive algorithms that analyze student responses and dynamically adjust task difficulty, thereby facilitating efficient skill acquisition and personalized

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learning trajectories [32]. Furthermore, AI-based tools allow learners to practice independently without the need for constant instructor supervision or physical instruments, increasing accessibility and flexibility in music training.

Related Research

Previous studies have demonstrated that technology-enhanced music instruction can positively influence student performance and learning outcomes. For instance, computer-assisted instruction has been associated with improvements in musical hearing, dictation, and performance skills [33–36]. Similarly, the integration of digital and mobile learning tools in music education has been shown to enhance student attitudes, engagement, and motivation. Previous studies have suggested that technology-enhanced and AI-supported learning environments may contribute to improvements in music-related learning processes by providing structured practice opportunities, adaptive feedback, and individualized instruction [37–41]. However, evidence regarding their effectiveness in undergraduate solfège education remains limited.

In the Turkish higher music education context, previous studies have increasingly emphasized the importance of technology-assisted instruction in music theory and solfège education. Gürer and Bilgin [34] reported that digital music theory platforms may enhance student engagement and support individualized practice opportunities in ear-training courses [38]. Özgül demonstrated that mobile-based music notation and training applications can contribute positively to students' learning motivation and accessibility in music education [41]. In addition, Çiçek highlighted the increasing academic interest in music theory and technology-oriented educational approaches in Türkiye [39]. Despite these developments, studies specifically investigating AI-supported solfège instruction in undergraduate conservatory education remain limited.

Another important aspect of technology integration in education is its influence on students' affective characteristics, such as attitudes toward learning. Attitude is a key determinant of learning success, as it reflects learners' motivation, engagement, and willingness to participate in instructional activities. Studies have indicated that technology-supported learning environments can foster positive attitudes by increasing interactivity, providing immediate feedback, and reducing performance anxiety [42–44]. In music education, where performance-related stress and self-confidence play crucial roles, AI-based systems may contribute to a more supportive and less intimidating learning environment by allowing students to practice privately and at their own pace.

Despite the growing interest in artificial intelligence-supported applications in education, there remains a significant gap in the literature regarding their pedagogical integration into undergraduate solfège instruction in real classroom settings. Traditional solfège education often provides limited individualized feedback, insufficient repetitive practice opportunities, and restricted adaptive learning support, particularly in higher music education environments with diverse learner needs. However, few studies have simultaneously examined both cognitive and affective learning outcomes within authentic classroom-based solfège instruction.

The present study assumed that adaptive intelligent tutoring systems may support individualized and self-regulated learning when integrated into structured instructional environments. However, the study is limited by its single-group design, institutional context, and relatively short intervention duration.

Accordingly, the present study was designed to address both the pedagogical and technological dimensions of artificial intelligence-supported solfège instruction in higher music education. Therefore, this study aimed to investigate the effects of artificial intelligence-supported solfège instruction on students' performance and learning attitudes within an action research framework. By integrating quantitative and qualitative data collected throughout a structured intervention process, the study seeks to provide a comprehensive evaluation of the pedagogical effectiveness of AI-supported learning in higher music education and to offer practical insights for technology-integrated solfège instruction.

2. Materials and Methods

2.1. Study Design

This study was conducted at the Department of Voice Education, Turkish Music Conservatory, Gaziantep University, Türkiye. This study was designed as action research employing a critical and developmental approach. Action research is a systematic, cyclical process in which practitioners actively participate in identifying problems, implementing interventions, and evaluating outcomes within real educational settings [45,46]. It aimed not only to generate knowledge but also to improve instructional practices and learning environments through iterative reflection and intervention.

In the present study, the first author acted as the instructor, while all authors contributed to the planning, monitoring, evaluation, and refinement of the research process. The research followed a structured action research cycle consisting of problem identification, planning, implementation, data collection, evaluation, and revision phases. This approach allowed for continuous monitoring and improvement of the AI-supported solfège instruction process. Throughout the intervention process, instructional practices were continuously reviewed according to student performance data, engagement patterns, classroom observations, and reflective notes recorded by the first author. Based on these observations, minor instructional modifications were implemented during the intervention period, including adjustments in exercise repetition frequency, task difficulty progression, pacing of weekly activities, and emphasis on specific auditory skill components requiring additional practice. This iterative plan–act–observe–reflect cycle was maintained throughout the implementation process in accordance with the principles of action research methodology.

2.2. Participants

The study was conducted with 36 undergraduate students enrolled in the Department of Voice Education at a conservatory in Türkiye during the 2023–2024 academic year. The participants were selected using a convenience sampling method because the intervention was conducted within the first author's teaching context.

The participants were aged between 20 and 39 years, with a mean age of approximately 22 years. The sample consisted of 72% female (n = 26) and 28% male (n = 10) students. Most participants (78%) were taking the solfège course for the first time, while a minority (22%) had previously repeated the course. All participants had access to smartphones, and the majority reported moderate to high levels of technological proficiency, which was essential for participation in the AI-supported learning environment (Table 1).

Table 1. Demographic Characteristics of Participants (n = 36).

Variable	Category	n	%
Gender	Female	26	72.2
	Male	10	27.8
Age (years)	Mean ± SD	22 ± 4.9	—
	Range	20–39	—
Course Status	First-time enrollment	28	77.8
	Repeating the course	8	22.2
Smartphone Ownership	Yes	36	100
	No	0	0

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Table 1. *Cont.*

Variable	Category	n	%
Computer Ownership	Yes	12	33.3
	No	24	66.7
Technological Proficiency	Moderate–High	35	97.2
	Low	1	2.8
Mother's Education Level	Primary education	21	58.3
	Secondary or higher	15	41.7
Father's Education Level	Primary education	15	41.7
	Secondary or higher	21	58.3
Parental Status	Living together	30	83.3
	Other	6	16.7
Number of Siblings	Mean	4	—
	Range	1–9	—

2.3. Intervention Procedure

In the context of the present study, the instructional framework was based on an intelligent tutoring system integrating adaptive learning, individualized feedback, performance monitoring, and repetitive auditory skill training. These pedagogical and technological components constituted the preliminary instructional environment underlying the intervention process. The intervention comprised a structured 10-week AI-supported solfège training program implemented through the EarMaster 7 software (EarMaster ApS, Aarhus, Denmark), an intelligent tutoring system specifically developed for music education. EarMaster integrates adaptive learning algorithms, real-time performance analysis, and individualized feedback mechanisms, enabling learners to engage in personalized and self-paced practice. The system functions as an intelligent tutoring system that employs rule-based adaptive instructional mechanisms rather than generative artificial intelligence models. Its core functions include response analysis, adaptive difficulty adjustment, performance tracking, and immediate corrective feedback. Throughout the intervention, weekly instructional content was systematically designed by the first author in consultation with the co-authors and delivered through the platform according to students' performance levels and learning progress. The training modules included core components of solfège education, namely interval recognition and production, solfège reading exercises, rhythmic training, and scale-based tonal exercises.

The instructional content was progressively modified in terms of complexity and scope based on students' performance data, in line with the principles of adaptive learning systems, which aim to optimize learning efficiency by aligning task difficulty with individual proficiency levels [32]. Participants were required to complete assigned weekly tasks; however, they were also encouraged to engage in additional independent practice beyond the minimum requirements to reinforce skill acquisition. The EarMaster system continuously tracked user activity, accuracy rates, task completion, and engagement patterns, thereby enabling real-time monitoring of student progress and facilitating data-driven instructional adjustments throughout the intervention period.

The overall instructional approach consisted of a structured AI-supported learning framework combining instructor-guided theoretical instruction with individualized adaptive practice through the EarMaster platform. All students received the same core instructional modules; however, exercise difficulty and repetition intensity were adaptively adjusted according to individual learner performance, accuracy levels, and progression patterns recorded by the system. The intervention process was standardized across the 10-week implementation period to ensure methodological consistency and reproducibility.

During the intervention period, students participated in weekly AI-supported solfège practice sessions consisting of approximately 3–4 structured exercises per week, with an average recommended practice duration of 30–45 min per session. All participants received the same core instructional content and weekly learning objectives; however, the adaptive system modified exercise difficulty, repetition frequency, and progression pace according to individual learner performance, response accuracy, and task completion patterns recorded by the platform. The first author continuously monitored student progress, reviewed engagement data, provided instructional guidance when necessary, and implemented minor pedagogical adjustments throughout the intervention process. These adjustments primarily involved pacing modifications, additional repetition of specific auditory skill exercises, and individualized reinforcement for students experiencing learning difficulties.

2.4. Data Collection Instruments

Both quantitative and qualitative data collection methods were used to ensure a comprehensive evaluation of the intervention. The evaluation framework of the study included pre-test, post-test, and delayed retention assessments together with continuous qualitative and system-generated monitoring throughout the intervention process. Quantitative evaluation focused on changes in attitude and performance outcomes, whereas qualitative evaluation examined learner experiences, engagement patterns, perceived benefits, and instructional challenges associated with the AI-supported learning environment.

Solfège Attitude Scale: The Solfège Attitude Scale was developed by the authors as a five-point Likert-type instrument to assess students' attitudes toward solfège instruction. The scale consisted of 26 items and was constructed in accordance with established principles of attitude measurement and scale development [47,48].

The development process included item pool generation based on relevant literature and student input, expert validation, pilot testing, and exploratory factor analysis to ensure construct validity. The final scale demonstrated high internal consistency, with a Cronbach's alpha coefficient of 0.96, indicating excellent reliability (Table 2).

Table 2. Psychometric Properties of the Solfège Attitude Scale.

Property	Result
Number of Items	26
Scale Type	5-point Likert
Initial Item Pool	167 items
Pilot Scale Items	60 items
Final Scale Items	26 items
Sample Size (Pilot Study)	324
KMO (Kaiser–Meyer–Olkin)	0.96
Bartlett's Test of Sphericity	$p < 0.001$
Number of Factors	2
Total Variance Explained (%)	63.1
Factor 1 (Positive Items)	13 items (0.31–0.83)
Factor 2 (Negative Items)	13 items (0.20–0.75)
Cronbach's Alpha (Total Scale)	0.96
Cronbach's Alpha (Factor 1)	0.95
Cronbach's Alpha (Factor 2)	0.94
Item Discrimination Analysis	$p < 0.05$

In addition, a Solfège Performance Test was developed to evaluate students' practical skills in solfège. The test comprised three core components: interval recognition, interval production, and solfège reading. It included a total of 27 items and was designed in alignment with course learning objectives and refined through expert consultation. Student performance was assessed using a binary scoring system (observed/not observed), which is widely utilized in performance-based evaluation to ensure objective and standardized assessment of skill acquisition [24] (Table 3).

Table 3. Structure of the Solfège Performance Test.

Component	Description	Number of Items	Scoring Method
Interval Recognition	Identification of musical intervals by auditory perception	9	Observed (1)/Not observed (2)
Interval Production	Vocal reproduction of given musical intervals	9	Observed (1)/Not observed (2)
Solfège Reading	Sight-reading and vocal performance of solfège exercises	9	Observed (1)/Not observed (2)
Total	—	27 items	—

Semi-Structured Interviews: Semi-structured interviews were conducted with participants before and after the intervention to explore their experiences, expectations, and perceptions of AI-supported learning. Interviews lasted between 30 and 50 min and were audio-recorded, transcribed, and analyzed. This method allowed in-depth exploration of student perspectives and is widely used in qualitative educational research [49].

Student Diaries: Students were asked to maintain weekly diaries throughout the intervention. These diaries included reflections on their learning experiences, challenges, and progress. Diary data provided valuable insights into students' emotional and cognitive engagement with the AI-supported learning process and supported triangulation of findings [50].

Researcher Diary: The first author maintained a reflective diary documenting observations, instructional decisions, and student responses throughout the intervention. This diary supported the interpretation of findings and enhanced the credibility of the research process [51].

System-Generated Data: The EarMaster software generated detailed learning analytics that enabled objective monitoring of student engagement and performance throughout the intervention. These system-generated data included time spent on tasks, accuracy rates, and progression levels across instructional modules. Such analytics provided continuous insight into students' learning behaviors and allowed for real-time tracking of skill development. The use of system-generated learning data is widely recognized as a key component of intelligent tutoring systems, facilitating both formative assessment and adaptive instructional design [32]. In the present study, these data were utilized to support both quantitative analyses (performance outcomes) and qualitative interpretations (learning processes and engagement patterns), thereby contributing to a comprehensive evaluation of the AI-supported instructional environment.

2.5. Data Collection Procedure

Data collection was carried out over a total period of 12 weeks, consisting of an initial two-week phase for pre-test administration and baseline interviews, followed by a 10-week intervention period. During the intervention, students participated in structured AI-supported solfège training while their progress and engagement were continuously monitored. Pre-test and post-test assessments were conducted to evaluate changes in both attitude and performance variables. In order to assess the sustainability of learning outcomes, a retention test was administered approximately 11 weeks after the completion of the intervention. This delayed measurement enabled the evaluation of long-term skill retention and provided additional insight into the persistence of the effects of AI-supported instruction (Figure 1).

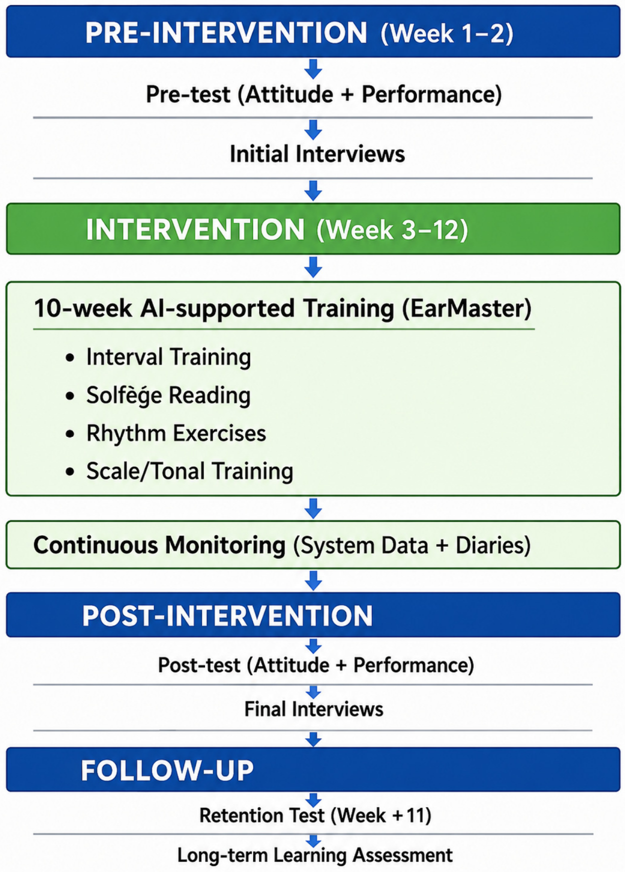

Figure 1. Flowchart of the action research process and intervention timeline.

2.6. Statistical Analysis

Quantitative data were analyzed using IBM SPSS Statistics (Version 27.0). The normality of the data distribution was assessed using the Shapiro–Wilk and Kolmogorov–Smirnov tests, which indicated that the data did not follow a normal distribution ($p < 0.05$). Accordingly, non-parametric statistical methods were employed. The Wilcoxon signed-rank test was used to compare pre-test and post-test scores for both attitude and performance variables, with statistical significance set at $p < 0.05$. Qualitative data obtained from interviews and student diaries were analyzed using content analysis, a systematic approach that involves coding data, identifying themes, and interpreting relationships between categories [9]. The analysis process consisted of four sequential stages: data coding, theme identification, organization of codes and themes, and interpretation of findings. This analytical framework enabled the identification of key patterns related to student experiences, engagement, and perceptions of AI-supported learning.

To improve the reliability and transparency of the qualitative analysis, interview transcripts, student diaries, and reflective notes were independently reviewed multiple times by the authors during the coding process. Initial codes were generated through open coding procedures and subsequently grouped into broader conceptual themes through iterative comparison and discussion among the authors. Representative participant quotations were included to support thematic consistency and enhance interpretive credibility. In addition, triangulation across interviews, student diaries, researcher reflections, and system-generated observations was used to strengthen the trustworthiness and dependability of the qualitative findings.

To ensure methodological rigor and trustworthiness, multiple validation strategies were implemented. These included the use of multiple data sources (triangulation), expert validation of measurement instruments, continuous observation and documentation throughout the intervention process, and the incorporation of participant feedback. Collectively, these procedures enhanced the credibility, dependability, and validity of the findings, which are fundamental quality criteria in action research studies. In addition to statistical comparisons, the evaluation process incorporated triangulation of quantitative findings, qualitative observations, and system-generated learning analytics in order to provide a comprehensive assessment of the instructional effectiveness and pedagogical impact of the intervention.

3. Results

The comparison of pre-test and post-test attitude scores revealed a statistically significant increase following the intervention ($Z = -3.58$, $p < 0.001$). The mean attitude score increased from 104.6 in the pre-test to 117.3 in the post-test. These findings indicate that the AI-supported instructional process contributed positively to students' learning attitudes and engagement toward solfège education (Table 4, Figure 2).

Table 4. Comparison of Pre- and Post-Test Attitude Scores.

Variable	Pre-Test Mean	Post-Test Mean	Z	*p*-Value
Attitude Score	104.6	117.3	−3.58	<0.001

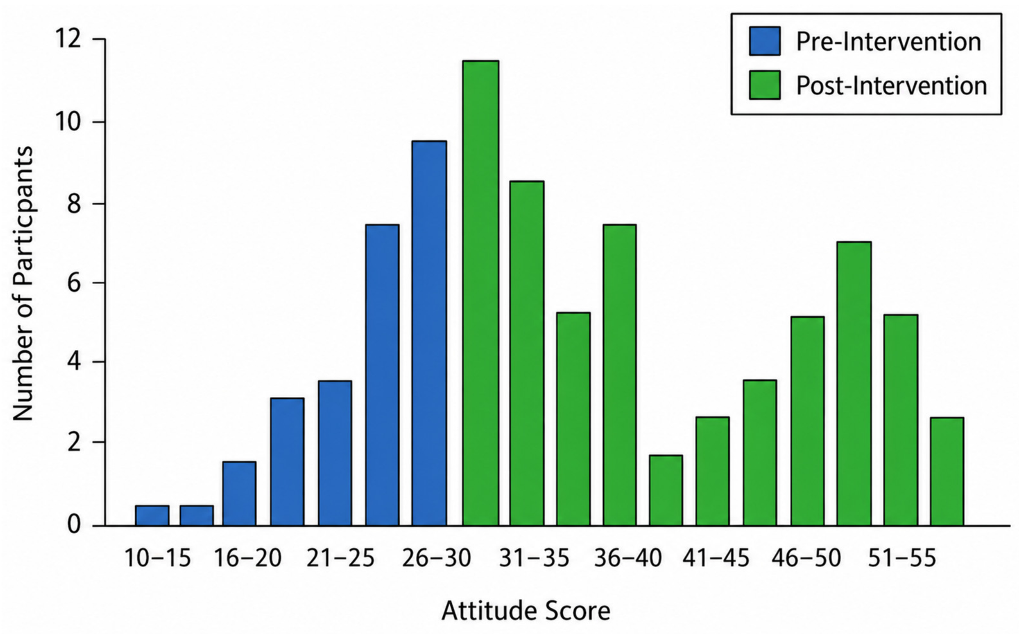

Figure 2. Distribution of Attitude Scores Before and After Intervention.

The comparison of pre-test and post-test performance scores revealed a statistically significant improvement following the intervention ($Z = 5.24$, $p < 0.001$). The mean performance score increased from 32.8 in the pre-test to 51.52 in the post-test. The observed improvement in performance scores suggests that adaptive practice and immediate feedback mechanisms may have facilitated auditory skill acquisition and performance development (Table 5, Figure 3).

Table 5. Comparison of Pre- and Post-Test Performance Scores.

Variable	Pre-Test Mean	Post-Test Mean	Z	*p*-Value
Performance Score	32.8	51.52	5.24	<0.001

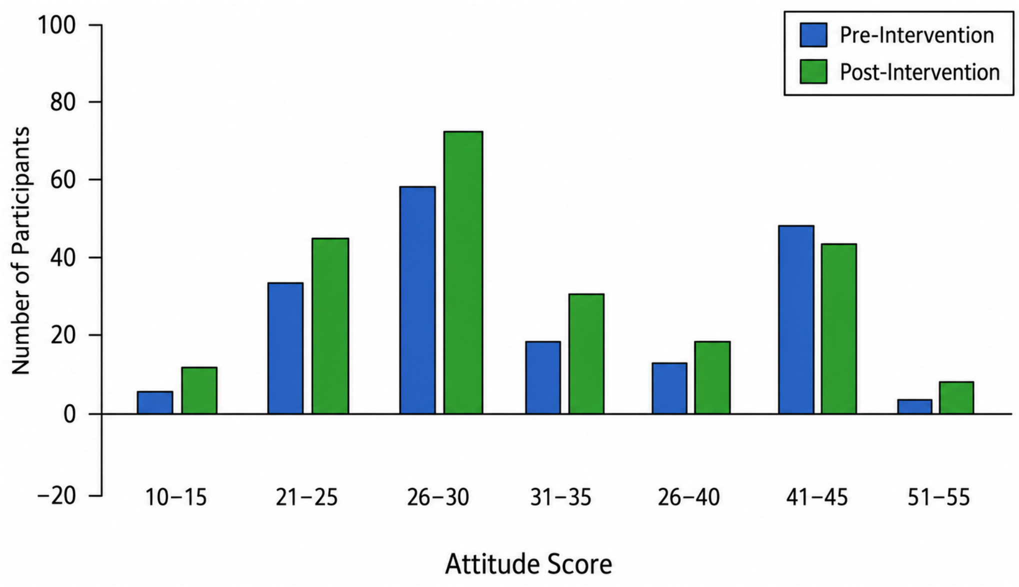

Figure 3. Changes in Performance Scores Across Subdomains.

Subgroup analysis of performance test components revealed statistically significant improvements across all domains following the intervention. Interval recognition (Z = 5.26, $p < 0.001$), interval production (Z = 5.25, $p < 0.001$), and solfège reading (Z = 5.27, $p < 0.001$) all demonstrated significant increases. The consistent improvement across all performance domains demonstrates that the intervention supported multiple dimensions of solfège competency rather than isolated skill development (Table 6).

Table 6. Subgroup Analysis of Performance Test Components.

Component	Z	*p*-Value
Interval Recognition	5.26	<0.001
Interval Production	5.25	<0.001
Solfège Reading	5.27	<0.001

A comparison of post-test and retention test scores revealed a statistically significant decline in performance following the no-practice period (Z = −4.25, $p < 0.001$). The mean performance score decreased from 51.5 in the post-test to 48.8 in the retention test. The decline observed during the retention phase suggests that continuous engagement and repetitive practice may be necessary to sustain long-term learning outcomes (Table 7, Figure 4).

Table 7. Comparison of Post-Test and Retention Test Scores.

Variable	Post-Test Mean	Retention Test Mean	Z	*p*-Value
Performance Score	51.5	48.8	−4.25	<0.001

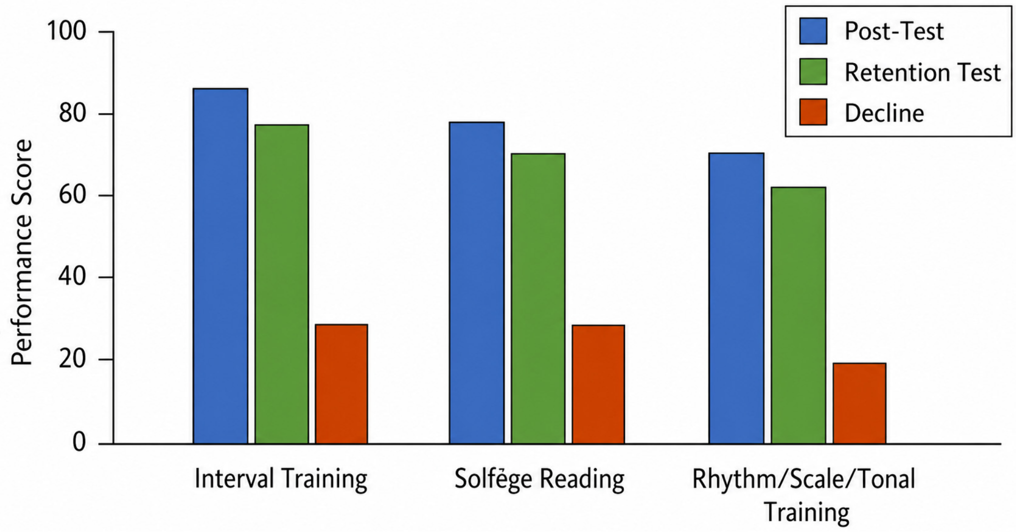

Figure 4. Performance Decline in Retention Test.

Analysis of performance changes according to engagement level indicated that students with higher levels of participation and consistent interaction with the AI-supported system demonstrated greater improvements in performance outcomes. In contrast, students with lower engagement levels exhibited more limited gains. These findings further highlight the importance of active learner participation and regular interaction with AI-supported instructional systems (Table 8).

Table 8. Performance Changes According to Engagement Level.

Engagement Level	Description	Observed Performance Change
High Engagement	Regular and frequent interaction with the system (e.g., daily participation)	Marked improvement
Moderate Engagement	Intermittent participation	Moderate improvement
Low Engagement	Irregular participation and limited interaction	Limited improvement

Qualitative analysis revealed several key themes related to the use of AI-supported solfège instruction. Students reported improvements in learning outcomes, increased engagement and motivation, and enhanced opportunities for personalized learning. The flexibility of time and location emerged as a major advantage, while reduced anxiety and increased confidence were noted as important psychological benefits. However, some challenges were also identified, including technical issues and the perceived workload associated with continuous practice. Overall, the qualitative findings supported the quantitative results by demonstrating increased motivation, confidence, and perceived learning effectiveness among participants (Table 9).

Table 9. Summary of Qualitative Themes and Representative Quotes.

Theme	Subtheme	Description	Representative Quote
Learning Outcomes	Skill Development	Improvement in auditory perception and solfège performance	"I feel that my interval recognition and solfège reading have significantly improved."
	Practice Efficiency	Increased effectiveness of repeated practice	"Practicing every day helped me understand the topics more clearly."

Table 9. *Cont.*

Theme	Subtheme	Description	Representative Quote
Engagement and Motivation	Enjoyment	Learning perceived as enjoyable and game-like	"It felt like a game, and I enjoyed practicing regularly."
	Motivation	Increased willingness to practice	"Seeing my progress motivated me to use the program more."
Personalized Learning	Individual Pace	Self-paced learning opportunities	"I could practice at my own pace without pressure."
	Immediate Feedback	Real-time correction of mistakes	"The system showed my mistakes instantly, which helped me improve."
Accessibility and Flexibility	Time Flexibility	Learning independent of time constraints	"I could study anytime, even outside class hours."
	Location Flexibility	Learning independent of physical setting	"I could practice anywhere without needing a classroom."
Psychological Impact	Reduced Anxiety	Lower performance anxiety compared to classroom	"I felt more comfortable practicing alone without being judged."
	Confidence	Increased self-confidence	"I became more confident in reading and performing solfège."
Challenges	Technical Issues	Problems related to software or devices	"Sometimes the system did not recognize my voice correctly."
	Workload	Perceived effort required	"The program required a lot of effort and time."

4. Discussion

The present study investigated the effects of AI-supported solfège instruction on students' performance and attitudes within an action research framework. The findings demonstrated that AI-supported instruction significantly improved both performance and attitudes, although a decline in performance was observed during the retention period. These results provide important insights into the pedagogical potential and limitations of artificial intelligence in music education.

One of the key findings of this study is the significant improvement in students' attitudes toward solfège instruction following the intervention. This finding is consistent with previous studies reporting that technology-enhanced learning environments positively influence students' attitudes and motivation. Andaç et al. and Şen et al. demonstrated that technology integration in music education enhances students' engagement and attitudes toward the course [52,53]. Özgül et al. reported that digital and mobile applications in solfège instruction contribute to more positive learner perceptions [41].

The positive shift in attitudes observed in this study can be attributed to several characteristics of AI-supported learning environments, including flexibility, interactivity, and immediate feedback. As highlighted by Wei et al., AI-based systems improve student engagement by offering personalized learning experiences and adaptive content delivery [43]. In addition, Lv et al. emphasized that AI-supported learning environments reduce cognitive load and increase learner satisfaction, which may explain the increased positive attitudes observed in this study [42].

However, an important finding of this study is that a subset of students exhibited a decline in attitude scores despite improvements in performance. This paradox suggests that AI-supported learning environments may impose increased cognitive and effort-related demands on learners. As discussed in the study findings, continuous feedback and the necessity for repeated practice may lead to frustration or fatigue in some students. Essel et al. reported that AI-supported learning systems, while effective, may increase perceived workload and cognitive strain [13]. Therefore, although AI-supported instruction enhances attitudes overall, educators should consider balancing system-driven feedback with motivational support to prevent potential negative affective responses.

The results demonstrated a substantial improvement in students' solfège performance following the AI-supported intervention. This finding is strongly supported by previous literature indicating that technology-enhanced and AI-supported learning environments improve performance outcomes in music education. Baurer et al. and Aydin et al. reported that computer-assisted instruction enhances musical performance skills, while Özgül et al. found improvements in melodic dictation performance among university students [41,54,55].

More specifically, Özdemir et al. demonstrated that the use of EarMaster significantly improves interval recognition and vocalization skills, which directly supports the findings of the present study [56]. Li et al. and Chen et al. emphasized that intelligent tutoring systems enhance performance by providing structured practice and real-time feedback [16,17].

The improvement across all subdomains (interval recognition, interval production, and solfège reading) further strengthens the argument that AI-supported instruction offers a comprehensive learning advantage. Yang et al. reported that AI-based music education systems support multidimensional skill development through adaptive learning mechanisms [33].

The observed improvements can be explained by several factors. First, AI-supported systems promote repeated practice, which is critical for skill acquisition in auditory-based disciplines. Second, immediate feedback allows learners to correct errors in real time, reinforcing correct responses. Third, adaptive learning ensures that tasks are aligned with individual skill levels, optimizing learning efficiency. These mechanisms are consistent with the principles of intelligent tutoring systems described by Guo et al. [32].

Another important finding of this study is the relationship between student engagement and performance improvement. Students who engaged more frequently with the AI-supported system demonstrated greater performance gains, while those with lower engagement showed limited improvement. This finding is consistent with previous research emphasizing the importance of active participation in AI-supported learning environments. Wei et al. [39] reported that the effectiveness of AI-based instruction is strongly dependent on learner interaction and engagement levels [43]. Essel et al. highlighted that consistent usage of AI-supported tools enhances learning outcomes through sustained practice [14]. The qualitative findings of this study also support this relationship, indicating that daily interaction with the system contributed to improved performance and confidence. This suggests that AI-supported instruction is not inherently effective but becomes effective when combined with consistent learner engagement.

Despite the significant improvements observed in post-test performance, a decline in retention scores was identified after the no-practice period. This finding highlights an important limitation of AI-supported instruction: the lack of sustained learning without continued engagement. This result is consistent with previous findings in skill-based learning domains. Özdemir et al. emphasized that continuous practice is essential for maintaining auditory skills in music education [56]. Kalkanoğlu et al. reported that the long-term effectiveness of technology-supported learning depends on ongoing practice and reinforcement [40]. The observed decline suggests that AI-supported systems should not be viewed as one-time interventions but rather as continuous learning tools. Integrating such systems into long-term curricula and encouraging sustained usage may enhance retention outcomes.

Nevertheless, the findings of the present study should be interpreted in light of several methodological limitations. The study employed a single-group pre-test/post-test action research design without a separate control group receiving traditional instruction. Therefore, potential confounding factors such as natural learning progression, repeated practice effects, increased familiarity with assessment procedures, and the Hawthorne effect cannot be fully excluded. Accordingly, the observed improvements should not be interpreted as being solely attributable to the AI-supported instructional system itself. Rather, the findings may reflect the combined influence of adaptive practice, increased engagement, structured instructional support, and continuous learner participation throughout the intervention process.

The qualitative findings provide valuable insights into the mechanisms underlying the observed improvements. Students reported increased motivation, enjoyment, and confidence, as well as reduced anxiety during learning. These findings are consistent with previous studies indicating that AI-supported environments create less stressful and more engaging learning experiences [28,42]. In particular, the reduction of performance anxiety appears to be a critical factor. As noted in the study findings, students felt more comfortable practicing independently without fear of judgement. Andaç et al. reported that technology-supported learning environments reduce performance-related anxiety [52]. Furthermore, the flexibility of time and location, as well as the ability to practice without physical instruments, were identified as key advantages. These findings are supported by Mazlan et al., who emphasized accessibility and flexibility as major benefits of AI-based music education systems [28]. The consistency observed across interviews, student diaries, reflective observations, and system-generated engagement patterns further strengthened the credibility of the qualitative findings.

The findings of this study have several important implications for music education. First, AI-supported systems such as EarMaster can significantly enhance both performance and attitudes when integrated effectively into the curriculum. Second, sustained engagement and practice are essential for maximizing learning outcomes. Third, educators should combine AI-supported tools with traditional teaching methods to create a balanced and effective learning environment.

In practical classroom settings, AI-supported systems such as EarMaster may be integrated into blended instructional models in which teacher-led instruction and individualized AI-based practice complement each other. For example, instructors may introduce theoretical concepts, rhythmic structures, and melodic exercises during face-to-face lessons, while AI-supported platforms can be used for individualized repetition, interval training, sight-reading practice, and formative assessment outside classroom hours. In this approach, the instructor maintains responsibility for pedagogical guidance, emotional support, interpretation of musical expression, and corrective feedback requiring human judgment, whereas the AI system provides repetitive practice opportunities, adaptive difficulty adjustment, immediate feedback, and continuous performance monitoring. Furthermore, system-generated learning analytics may assist instructors in identifying students with learning difficulties, monitoring engagement patterns, and adapting instructional content according to individual learner needs. Such a hybrid instructional model may enhance both instructional efficiency and personalized learning opportunities in higher music education.

Despite its advantages, some challenges associated with AI-supported instruction were identified. These include technical issues, increased workload, and unequal access to technological resources. Additionally, although AI systems provide effective feedback, they cannot fully replace the role of human instructors. As emphasized in the study findings, teacher support remains essential for addressing emotional and motivational aspects of learning.

The findings of this study should also be interpreted within the broader context of pedagogical systems and learner-centered instructional design. AI-supported learning environments may contribute to the transformation of traditional teacher-centered music instruction into more adaptive, participatory, and self-regulated learning processes. Rather than functioning solely as technological support tools, intelligent tutoring systems may facilitate individualized learning pathways, continuous formative assessment, and reflective learning experiences. In this context, the role of the instructor evolves from direct knowledge transmission toward guidance, facilitation, motivational support, and pedagogical supervision. Accordingly, the effectiveness of AI-supported instruction may depend not only on the technological capabilities of the system itself, but also on how effectively it is pedagogically integrated into the teaching and learning environment.

Several open research questions remain regarding the long-term pedagogical, socio-technical, and ethical implications of artificial intelligence-supported music education. First, future studies should investigate how sustained interaction with intelligent tutoring systems influences long-term retention, self-regulated learning behaviors, and creative musical development across different educational levels. Second, questions remain regarding equitable access to technology-supported learning environments, particularly for students with limited technological resources or differing levels of digital literacy. In addition, although adaptive instructional systems may improve individualized learning opportunities, excessive reliance on technology may potentially reduce direct social interaction and instructor–student engagement in music education contexts. Ethical considerations related to student data monitoring, learning analytics, algorithmic transparency, and digital dependency should also be examined more comprehensively in future research. Furthermore, comparative studies involving different instructional models, larger multicenter samples, and longer intervention periods are needed to better understand the broader educational impact and sustainability of artificial intelligence-supported solfège instruction.

5. Conclusions

In conclusion, this study demonstrated that artificial intelligence-supported solfège instruction positively influenced students' performance and learning attitudes in higher music education. The findings suggest that adaptive feedback, individualized practice, and continuous learner engagement may contribute to improved auditory skill development and student motivation. However, sustained practice appears necessary for long-term retention of learning outcomes. Overall, artificial intelligence-supported instructional systems may serve as effective complementary tools when integrated with teacher-guided music education.

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